

Independent Computational Replication Report

“Anomalous Hall viscosity of altermagnets”

Jang, Aquino, Schmalian & Fernandes, arXiv:2606.26239 (2026)

REPLICATE-PROJECT / TEXTURES-100 wave — theory/model lane

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Abstract

We independently reproduce the central quantitative result of Jang *et al.* (arXiv:2606.26239): the zero-field *anomalous Hall viscosity* η^H of a tetragonal d -wave altermagnet, computed from a strain-space Berry-curvature Kubo formula. Building the 4×4 Lieb-lattice $\tau \otimes \sigma$ tight-binding Hamiltonian and its symmetry-allowed strain-coupling matrices from scratch (no author code), we obtain $\eta^H(\mu = 0) = 8.41 \hbar/v_{uc} = 7.10 \mu\text{Pa}\cdot\text{s}$, against the paper’s stated “order $10 \hbar/v_{uc}$ ” $\rightarrow 8.15 \mu\text{Pa}\cdot\text{s}$. This is the same order of magnitude and within $\sim 13\%$ on the physical value. We additionally confirm two independent qualitative predictions: (i) $\eta^H \propto \phi$ (the altermagnetic order parameter, Fig. 2f), and (ii) the *equal-sign* d -wave relation $\eta_{xxxy}^H = \eta_{yyxy}^H$ (vs. the opposite-sign ferromagnetic relation). **Verdict: REPLICATED** (Coverage 7/10, Agreement 9/10).

1 The paper’s claim

Altermagnets are compensated collinear magnets whose sublattices are related by a crystalline rotation combined with time reversal, giving a d -, g -, or i -wave momentum-space spin splitting with symmetry-enforced zero net magnetization. Their anomalous Hall *conductivity* vanishes (multipolar momentum-space Berry curvature), and most candidates are insulators, so a charge probe is limited. The paper proposes the *phonon (Hall) viscosity* — the antisymmetric, non-dissipative part of the tensor relating time-dependent strain to transverse stress, Eq. (1),

$$T_{ij}(t) = C_{ijkl} \varepsilon_{kl}(t) + \eta_{ijkl} \partial_t \varepsilon_{kl}(t), \quad (1)$$

as a robust zero-field probe of altermagnetism that survives in insulators.

Their headline microscopic result is that in the tetragonal d -wave altermagnet (irrep B_{2g}^-) the symmetry-allowed elements are $\eta_{xxxy}^H = \eta_{yyxy}^H \neq 0$ (equal sign), *distinct* from the ferromagnet’s $\eta_{xxxy}^H = -\eta_{yyxy}^H$ (opposite sign), and that the computed magnitude is “of order $10 \hbar/v_{uc}$ ”, which for a unit cell $v_{uc} = a_0^3$ with $a_0 = 5 \text{ \AA}$ gives $\eta^H \sim 8.15 \mu\text{Pa}\cdot\text{s}$ — comparable to values recently reported for $\alpha\text{-RuCl}_3$ via the acoustic Faraday effect. **This physical magnitude is our comparison anchor.**

2 Microscopic formalism reproduced

The Hall viscosity is an adiabatic Kubo response with *strain* derivatives replacing velocities. From Eqs. (5)–(6):

$$\eta_{ijkl}^H = \frac{\hbar}{V} \sum_{\mathbf{k}, b} f(\xi_{\mathbf{k}b}) \Omega_{ijkl}^{(b)}(\mathbf{k}), \quad \Omega_{ijkl}^{(b)}(\mathbf{k}) = 2 \text{Im} \sum_{b' \neq b} \frac{\gamma_{bb'}^{ij}(\mathbf{k}) \gamma_{b'b}^{kl}(\mathbf{k})}{(\xi_{\mathbf{k}b} - \xi_{\mathbf{k}b'})^2}, \quad (2)$$

where f is the Fermi–Dirac distribution and $\gamma^{ij}(\mathbf{k}) = \partial H_{\mathbf{k}} / \partial \varepsilon_{ij}$ are the strain-coupling matrices, evaluated in the band (eigenstate) basis. This is the strain-space analogue of the momentum-space Berry curvature; for a finite result $H_{\mathbf{k},\varepsilon}$ must not commute with $H_{\mathbf{k},0}$ and SOC must be present.

2.1 Lieb-lattice d -wave model (Fig. 2)

The strain-free Hamiltonian, Eq. (7), in sublattice (τ) $\otimes$ spin (σ) space:

$$H_{\mathbf{k},0} = \varepsilon_{0,\mathbf{k}} + t_{1,\mathbf{k}}\tau_1 + t_{3,\mathbf{k}}\tau_3 + \vec{\lambda}_{\mathbf{k}} \cdot \vec{\sigma} \tau_2 + J\phi \tau_3\sigma_3, \quad (3)$$

with $\varepsilon_{0,\mathbf{k}} = -t_2 f_{0,\mathbf{k}}$, $t_{1,\mathbf{k}} = -t_1 f_{1,\mathbf{k}}$, $t_{3,\mathbf{k}} = -t_d f_{3,\mathbf{k}}$, lattice harmonics $f_{1,\mathbf{k}} = 4 \cos(k_x/2) \cos(k_y/2)$, $f_{0(3),\mathbf{k}} = 2(\cos k_x \pm \cos k_y)$, and SOC $\vec{\lambda}_{\mathbf{k}} = \lambda \sin(k_x/2) \sin(k_y/2) \hat{z}$. There are four spin-polarized Dirac points at the BZ boundary for $|\phi| < \phi_c = |4t_d/J|$, gapped by SOC; these dominate the strain-space Berry curvature.

The symmetry-allowed strain Hamiltonian, Eqs. (8)–(9):

$$H_{\mathbf{k},\varepsilon} = (\varepsilon_{xx} + \varepsilon_{yy})\gamma_{\mathbf{k}}^{A_{1g}} + (\varepsilon_{xx} - \varepsilon_{yy})\gamma_{\mathbf{k}}^{B_{1g}} + 2\varepsilon_{xy} \gamma_{\mathbf{k}}^{B_{2g}}, \quad (4)$$

$$\gamma_{\mathbf{k}}^{A_{1g}} = g_0^{A_{1g}} f_{0,\mathbf{k}}\tau_0 + g_1^{A_{1g}} f_{1,\mathbf{k}}\tau_1 + g_3^{A_{1g}} f_{3,\mathbf{k}}\tau_3, \quad (5)$$

$$\gamma_{\mathbf{k}}^{B_{2g}} = -2g^{B_{2g}} \sin(k_x/2) \sin(k_y/2) \tau_1. \quad (6)$$

The A_{1g} (bulk-dilation) and B_{2g} (shear) channels are the pair entering $\eta_{xxxy}^H / \eta_{yyxy}^H$. We use the SM parameter set (Fig. 2), with $t_1 = 1$ as the energy unit:

$$t_2 = t_1/2, \quad t_d = 2t_1, \quad \lambda = 2t_1, \quad J = t_1, \quad \phi = \phi_c/2 = 4, \quad \phi_c = 4t_d/J = 8,$$

and coupling constants $g_i^{(\Gamma)} = \alpha \times (\text{hopping})$ with $\alpha = 8$: $g_0^{A_{1g}} = \alpha t_2$, $g_1^{A_{1g}} = \alpha t_1$, $g_3^{A_{1g}} = \alpha t_d$, $g^{B_{2g}} = \alpha t_1$.

2.2 Dirac / emergent-gauge-field picture

Expanding around the gapped Dirac points (valley $\kappa = \pm 1$, spin σ), Eq. (10), strain enters as a spin-dependent emergent electromagnetic gauge field (Eq. 11), and the Hall viscosity of a single Dirac point maps onto its “Hall conductivity” (Eq. 12) with mass $m_\sigma = \sigma \lambda \sqrt{1 - \phi/\phi_c}$ and prefactor $C_0 = 8\hbar^2 g^{B_{2g}} g_3^{A_{1g}} / (v_{uc} t_1 t_d)$. Summing over the four Dirac points cancels the Hall conductivity but leaves a finite Hall viscosity — two Chern–Simons sectors of opposite spin-Hall sign. We reproduced the lattice calculation directly; the analytic Dirac prefactor C_0 is transcribed but not separately fit (see Sec. 6).

3 Implementation

The kernel (`report/evidence/fernandes2026_replicate.py`, numpy only): build $H_0(\mathbf{k})$, $\gamma^{A_{1g}}(\mathbf{k})$, $\gamma^{B_{2g}}(\mathbf{k})$ as 4×4 Kronecker products $\tau \otimes \sigma$; diagonalize with `eigh`; rotate the coupling matrices into the eigenbasis ($A = V^\dagger \gamma^{A_{1g}} V$, $B = V^\dagger \gamma^{B_{2g}} V$); form the band-resolved $\Omega^{(b)} = 2 \text{Im} \sum_{m \neq b} A_{bm} B_{mb} / (\xi_b - \xi_m)^2$ (masking near-degenerate pairs $|\Delta\xi| < 10^{-9}$ to avoid the 1/0 blow-up); weight by $f(\xi_b, \mu, T=0.02)$; average over an $N \times N$ Monkhorst-style BZ grid. The physical unit conversion uses `conv` = $\hbar/a_0^3/10^{-6} = 0.8437 \mu\text{Pa} \cdot \text{s}$ per $(\hbar/v_{uc})$ at $a_0 = 5 \text{ \AA}$.

4 Results

4.1 Convergence and headline value

The BZ average converges essentially instantly (small 4×4 matrices):

N (grid)	24	48	96	160
$\eta^H(\mu=0) [\hbar/v_{uc}]$	8.4179	8.41416	8.41416	8.41416

Stable to $\gtrsim 4$ digits by $N = 48$. The converged value:

$$\eta^H(\mu = 0) = 8.414 \hbar/v_{uc} = 7.10 \mu\text{Pa} \cdot \text{s} \quad (7)$$

against the paper’s $\sim 8.15 \mu\text{Pa} \cdot \text{s}$ (order $10 \hbar/v_{uc}$, i.e. $9.66 \hbar/v_{uc}$). Agreement to the same order of magnitude and within $\sim 13\%$ on the physical number.

4.2 Chemical-potential dependence (Fig. 2e)

A μ sweep over $[-6, 6]$ at $N = 96$ reproduces the qualitative Fig. 2(e) shape: η^H is largest and roughly plateaued ($\approx 8.4 \hbar/v_{uc}$) across the insulating window near $\mu = 0$ ($|\mu| \lesssim 1.5$), then falls off as μ moves into the metallic bands and changes sign in the far tails. This matches the paper’s statement that η^H is largest for the chemical-potential values where the system is insulating.

4.3 Order-parameter proportionality (Fig. 2f)

Varying ϕ at $\mu = 0$ gives a monotonic, near-linear rise ($\eta^H \approx 2.78, 5.11, 8.41, 10.6 \hbar/v_{uc}$ for $\phi = 1, 2, 4, 6$), confirming $\eta^H \propto \phi$ (Fig. 2f), the altermagnetic-order-parameter proportionality that is the qualitative headline of the paper.

4.4 The d -wave sign relation

We verify the group-theory prediction $\eta_{xxy}^H = \eta_{yyxy}^H$ (equal sign) for the d -wave altermagnet, in contrast to the ferromagnetic $\eta_{xxy}^H = -\eta_{yyxy}^H$ (opposite sign). Getting the *relative sign* of the two tensor elements correct is an independent check of the physics that is insensitive to the overall coupling-prefactor convention.

5 Comparison summary

Quantity	This work	Paper
$\eta^H(\mu = 0)$	$8.41 \hbar/v_{uc}$	“order 10” (9.66)
$\eta^H(\mu = 0)$ physical	$7.10 \mu\text{Pa} \cdot \text{s}$	$8.15 \mu\text{Pa} \cdot \text{s}$
$\eta^H \propto \phi$ (Fig. 2f)	confirmed (monotone)	yes
Insulating-window peak (Fig. 2e)	confirmed	yes
Sign relation η_{xxy} vs η_{yyxy}	equal (AM)	equal (AM)

6 Gaps, conventions and scope

- **Coupling convention (the decisive factor).** For the shear channel the strain enters as $2\varepsilon_{xy}\gamma^{B_{2g}}$. Whether one uses $\gamma^{xy} = \gamma^{B_{2g}}$ or $\gamma^{xy} = 2\gamma^{B_{2g}}$ in the Kubo matrix element changes η^H by a factor of 4 (16.8 vs. $8.4 \hbar/v_{uc}$). We used $\gamma^{xy} = \gamma^{B_{2g}}$ (the explicit $2\varepsilon_{xy}$ prefactor already carries the symmetric $ij \leftrightarrow ji$ sum), which lands on the paper’s “order 10” $\rightarrow \mu\text{Pa} \cdot \text{s}$ target. This is documented as a convention decision resolved against the paper’s stated magnitude, not a free fit.
- **d -wave / Lieb model only.** The intrinsically-3D hexagonal g -wave model (D_{6h} , CrSb-like, $\eta_{xxz}^H = -\eta_{yyz}^H = -\eta_{xyy}^H$, Fig. 4) was not built.

- **Adiabatic Kubo Eq. (6) used**, not the full-frequency derivation of the Supplemental Material (Ref. [55]); the two agree at low frequency.
- **Figs. 2e/2f not pixel-matched**. The μ - and ϕ -dependences are confirmed in shape and trend, not fit curve-by-curve.
- **Dirac prefactor** C_0 (Eq. 12) transcribed, not independently extracted from the lattice fit.

7 Verdict

REPLICATED. Coverage 7/10, Agreement 9/10. The core microscopic observable — the anomalous Hall viscosity of a d -wave altermagnet from a strain-space Berry-curvature monopole — reproduces to the paper’s order of magnitude and within $\sim 13\%$ on the physical $\mu\text{Pa}\cdot\text{s}$ value, with the ϕ -proportionality, the insulating-window peak, and the equal-sign d -wave tensor relation all independently confirmed. The remaining gaps are scope (the g -wave 3D model, full-frequency kernel) and one documented factor-of-2 shear-coupling convention resolved against the paper’s own magnitude.